MultiQT notes

November 11, 2026

1 Notation

Unless stated otherwise, fix one disease, location and forecast horizon.

Forecasts. The index t labels a forecast round (reference week), and y_t is the outcome for its target week. Use s for an earlier round. At quantile levels $\mathcal{A} = \{\alpha_1 < \cdots < \alpha_m\} \subset (0,1)$, b_t^α is the base forecast and q_t^α the recalibrated forecast. The vectors $b_t, q_t \in \mathbb{R}^m$ collect all levels. Values are hospital counts unless a transformation is shown.

Coverage and loss. Write $\mathbf{1}\{\cdot\}$ for an indicator and $x_+ = \max(x, 0)$. Coverage at level α is $\mathrm{cov}_t^\alpha = \mathbf{1}\{y_t \le q_t^\alpha\}$, with target α. Quantile loss, with prediction z first and observation r second, is

$$\rho_\alpha(z, r) = \alpha(r - z)_+ + (1 - \alpha)(z - r)_+. \tag{1}$$

Ordering and states. Forecasts must be nondecreasing across levels. Write

$$\mathcal{K} = \{q \in \mathbb{R}^m : q^{\alpha_1} \le \cdots \le q^{\alpha_m}\},$$

and assume $b_t \in \mathcal{K}$. Projection onto a nonempty closed convex set C means the nearest point in Euclidean distance:

$$\Pi_C(x) = \arg\min_{z \in C} \|z - x\|^2.$$

A tilde marks hidden accumulated state; the untilded state is its projected version. Standard offsets use $\tilde{\theta}_t, \theta_t$; studentized offsets use $\tilde{u}_t, u_t$. Superscripts $\mathrm{std}, \mathrm{stu}$ distinguish these methods, and η or η_t is the step size.

The methodological recursions forecast, observe y_t, then update. Delayed reports and revisions are introduced in Section 3.2; implementation, weighting and evaluation symbols are defined where they are used.

2 Methodological notes

2.1 Studentized score

Standard MultiQT tracks additive residual corrections. Studentization tracks scaled residuals, so changing the scale also changes the accumulated correction in outcome units. A falling scale can shrink a positive correction after a peak without erasing the hidden state. The quantile coverage target is unchanged; this construction alone does not establish calibration guarantees.

2.1.1 Standard MultiQT and score tracking

Starting with zero hidden state, MultiQT [1] projects the forecast but updates the unprojected state:

$$\theta_t = \Pi_{\mathcal{K}-b_t}(\tilde{\theta}_t), \qquad q_t^{\text{std}} = b_t + \theta_t, \tag{2}$$

$$\text{cov}_t^{\text{std},\alpha} = \mathbf{1}\{y_t \le q_t^{\text{std},\alpha}\}, \qquad \tilde{\theta}_{t+1}^{\alpha} = \tilde{\theta}_t^{\alpha} - \eta(\text{cov}_t^{\text{std},\alpha} - \alpha). \tag{3}$$

For a score $S_t(y)$, let $r_t = S_t(y_t)$ be its observed value, c_t its threshold and $\delta = 1 - \alpha$ the target miscoverage. The scalar score tracker of Angelopoulos et al. [2] uses

$$\mathcal{P}_t = \{y : S_t(y) \le c_t\}, \qquad \text{err}_t = \mathbf{1}\{r_t > c_t\}, \tag{4}$$

and updates

$$c_{t+1} = c_t + \eta(\text{err}_t - \delta) = c_t - \eta(\mathbf{1}\{r_t \le c_t\} - \alpha). \tag{5}$$

This is a quantile-loss subgradient step. In the threshold argument,

$$\partial_z \rho_\alpha(z, r) = \begin{cases} \{-\alpha\}, & z < r, \\ [-\alpha, 1-\alpha], & z = r, \\ \{1-\alpha\}, & z > r, \end{cases} \tag{6}$$

and the selected subgradient, including ties, is

$$g_\alpha(z; r) = \mathbf{1}\{r \le z\} - \alpha \;\in\; \partial_z \rho_\alpha(z, r). \tag{7}$$

Loss depends on distance from the threshold; the update depends only on which side contains the observation.

2.1.2 Residual coordinates

The signed residual score recovers standard MultiQT:

$$S_t^{\text{std},\alpha}(y) = y - b_t^{\alpha}, \qquad r_t^{\text{std},\alpha} = y_t - b_t^{\alpha}. \tag{8}$$

Its threshold θ_t^{α} defines the forecast endpoint and coverage:

$$\{y : S_t^{\text{std},\alpha}(y) \le \theta_t^{\alpha}\} = (-\infty, b_t^{\alpha} + \theta_t^{\alpha}] = (-\infty, q_t^{\text{std},\alpha}], \tag{9}$$

$$\text{cov}_t^{\text{std},\alpha} = \mathbf{1}\{r_t^{\text{std},\alpha} \le \theta_t^{\alpha}\}. \tag{10}$$

Thus

$$\tilde{\theta}_{t+1}^{\alpha} = \tilde{\theta}_t^{\alpha} - \eta(\mathbf{1}\{r_t^{\text{std},\alpha} \le \theta_t^{\alpha}\} - \alpha), \tag{11}$$

with the same loss in residual coordinates:

$$\ell_t^{\text{std}}(\theta) = \sum_{\alpha \in \mathcal{A}} \rho_\alpha(b_t^{\alpha} + \theta^{\alpha}, y_t) = \sum_{\alpha \in \mathcal{A}} \rho_\alpha(\theta^{\alpha}, r_t^{\text{std},\alpha}). \tag{12}$$

For one level there is no ordering constraint, so this is scalar quantile tracking. For several levels, evaluate the subgradient at the projected state and apply it to the hidden state.

2.1.3 Studentized residuals and their objective

Choose finite, positive scales σ_t^α from information available at the forecast cutoff and hold them fixed when differentiating. Offsets u_t^α are in studentized units, whereas θ_t^α is in outcome units. Define

$$S_t^{\mathrm{stu},\alpha}(y) = \frac{y - b_t^\alpha}{\sigma_t^\alpha}, \qquad r_t^{\mathrm{stu},\alpha} = \frac{y_t - b_t^\alpha}{\sigma_t^\alpha}. \tag{13}$$

The score is increasing, so inversion preserves the coverage event:

$$q_t^{\mathrm{stu},\alpha} = b_t^\alpha + \sigma_t^\alpha u_t^\alpha, \qquad \mathrm{cov}_t^{\mathrm{stu},\alpha} = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\}. \tag{14}$$

Quantile loss of the standardized scores is a weighted outcome-unit loss:

$$\ell_t^{\mathrm{stu}}(u) = \sum_{\alpha\in\mathcal{A}} \rho_\alpha(u^\alpha, r_t^{\mathrm{stu},\alpha}) = \sum_{\alpha\in\mathcal{A}} \frac{\rho_\alpha(b_t^\alpha + \sigma_t^\alpha u^\alpha, y_t)}{\sigma_t^\alpha}. \tag{15}$$

Its selected subgradient is

$$g_t^{\mathrm{stu},\alpha}(u_t) = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\} - \alpha = \mathrm{cov}_t^{\mathrm{stu},\alpha} - \alpha. \tag{16}$$

Indeed, away from ties the inverse scale in the loss cancels the scale from differentiation of the forecast:

$$\frac{\partial}{\partial u^\alpha} \frac{\rho_\alpha(b_t^\alpha + \sigma_t^\alpha u^\alpha, y_t)}{\sigma_t^\alpha} = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u^\alpha\} - \alpha. \tag{17}$$

At ties use (7). Starting from $\tilde{u}_1 = 0$, the update is

$$\tilde{u}_{t+1}^\alpha = \tilde{u}_t^\alpha - \eta g_t^{\mathrm{stu},\alpha}(u_t). \tag{18}$$

The projected threshold determines coverage; only the hidden threshold accumulates increments. This is not gradient descent on the unweighted outcome-unit loss in u: that gradient has an additional factor σ_t^α. The coding section also tests median-centred forecasts and nonzero initial states.

2.1.4 Ordering and projection

Ordering always concerns the returned forecasts. For standard residuals,

$$C_t^{\mathrm{std}} = \mathcal{K} - b_t, \qquad b_t + \theta \in \mathcal{K} \iff \theta \in C_t^{\mathrm{std}}, \tag{19}$$

$$\theta^{\alpha_i} - \theta^{\alpha_{i+1}} \le b_t^{\alpha_{i+1}} - b_t^{\alpha_i}, \qquad i = 1, \dots, m-1. \tag{20}$$

For studentized residuals, collect the scales in $A_t = \mathrm{diag}(\sigma_t^{\alpha_1}, \dots, \sigma_t^{\alpha_m})$. Then

$$C_t^{\mathrm{stu}} = \{u : b_t + A_t u \in \mathcal{K}\} = A_t^{-1}(\mathcal{K} - b_t). \tag{21}$$

This is a nonempty closed convex polyhedron containing zero, generally not a cone. Unequal scales do not preserve $\mathcal{K}$, so it cannot generally be written as $\mathcal{K} - A_t^{-1} b_t$. The unique Euclidean projection gives

$$u_t = \Pi_{C_t^{\mathrm{stu}}}(\tilde{u}_t), \qquad q_t^{\mathrm{stu}} = b_t + A_t u_t. \tag{22}$$

Changing variables to $q = b_t + A_t u$ yields weighted isotonic regression:

$$q_t^{\mathrm{stu}} = \arg\min_{q\in\mathcal{K}} \sum_{\alpha\in\mathcal{A}} \frac{(q^\alpha - b_t^\alpha - \sigma_t^\alpha \tilde{u}_t^\alpha)^2}{(\sigma_t^\alpha)^2}, \tag{23}$$

For positive-definite W, write $\Pi_{\mathcal{K}}^W(v) = \arg\min_{q\in\mathcal{K}} (q - v)^\top W (q - v)$ for weighted projection. Then the equivalent projection identity is

$$b_t + A_t \Pi_{A_t^{-1}(\mathcal{K} - b_t)}(\tilde{u}_t) = \Pi_{\mathcal{K}}^{A_t^{-2}}(b_t + A_t \tilde{u}_t). \tag{24}$$

Weights $1/(\sigma_t^\alpha)^2$ follow from Euclidean projection in score coordinates; ordinary residual coordinates give equal weights.

2.1.5 Common and constant scales

For a scale σ_t common to all levels, $A_t = \sigma_t I$, where I is the identity matrix. Positive scalar multiplication preserves $\mathcal{K}$:

$$C_t^{\text{stu}} = \mathcal{K} - b_t/\sigma_t, \qquad q_t^{\text{stu}} = \Pi_{\mathcal{K}}(b_t + \sigma_t \tilde{u}_t). \tag{25}$$

The projection is ordinary isotonic regression. This is standard MultiQT on b_t/σ_t and y_t/σ_t, with the returned forecast multiplied by σ_t. For constant σ, it is also equivalent to standard MultiQT on the original inputs after matching initialization and converting state and rate:

$$\tilde{\theta}_t = \sigma \tilde{u}_t, \qquad \eta_{\text{std}} = \sigma \eta_{\text{stu}}. \tag{26}$$

For varying σ_t, changing the scale rescales the accumulated state; changing only the current rate does not reproduce that effect.

2.2 Questions and ideas

Which property of the base forecast should another score recalibrate? Increasing, invertible scores preserve the quantile coverage event; changing the target to PIT or highest-density-region calibration would require a separate formulation.

3 Coding Updates

The tracker transform is $\psi(x) = x$ or $\log(1 + x)$, with transformed base $\beta_t^\alpha = \psi(b_t^\alpha)$. Offsets θ_t are in these tracker units. The count divisor c_{scale} used for scoring is population/100,000 by default.

3.1 Data

Script 00 downloads CMU–TimeSeries hub forecasts and dated NHSN reports, and generates our own base forecasts where the required inputs are available. Script 01 prepares cases (one disease, location and forecast horizon) from the selected base forecasts and reports. Cases retain generation cutoffs, revisions, feedback schedules for delays 0–4, populations and filled-forecast flags.

Reference weeks are Saturdays (`reference_date`); the target week of y_t is `target_end_date` in forecasts and `target_date` in prepared cases. The information cutoff ι_t is normally the Wednesday of the reference week, or the actual generation date where known. Write $y_s^{[v]}$ for the latest report for the target week of row s, dated on or before cutoff v (archive field `version`). A valid report is finite and nonnegative. "Latest counts" means the latest reports in the downloaded archive, through 30 October 2026.

Hub forecasts: the active design. The `hub_first_months` design uses submitted CMU–TimeSeries forecasts for both diseases from reference date 23 November 2024. The stored quantiles are used subject to the cleaning and filling below. Feedback uses the report available at the relevant cutoff, while evaluation uses latest counts.

Target dates determine the split: tuning through 31 May 2025, then evaluation from June 2025 through September 2026. Overall scores use all available evaluation weeks; before/after-peak scores restrict these to the corresponding peak windows. Flu has long summer gaps in submitted forecasts, rather than an explicit winter-only evaluation rule. The flu winter 2024/25 peak belongs to tuning. Plotted target weeks after September 2026 are outside the evaluation period and are not scored.

Own `windowed_seasonal` forecasts: the longer-history alternative. The longer-history setup starts on reference date 14 October 2023, with tuning target dates through 30 September 2024 and evaluation target dates from 1 October 2024 through 30 September 2026. Our COVID forecasts have been generated with unchanged, pinned Delphi code and packages. Flu still uses hub forecasts: we have not generated the corresponding own flu series. Earlier flu hub coverage is incomplete, especially at horizon -1; this is not a matched comparison of two locally generated base models. The output guide below uses the active hub design.

Why own flu forecasts are blocked. Flu adds historical ILI+ and adjusted FluSurv-NET training rows to the hospital counts; COVID uses hospital counts alone. Constructing these additional rows requires eight CSVs absent locally: `flu_burden.csv`, `us_pop.csv` and six WHO/NREVSS laboratory files. Script 00 skips flu without them. Public reconstruction sources exist, but matching the original input versions remains unverified. Omitting the additional rows would give a version of `windowed_seasonal` trained only on hospital data, with less history and potentially different forecasts. The published notebooks read prepared archives and saved forecasts; they do not grant access to the private data bucket. The exact inputs or a prepared training archive could be requested.

How earlier reports are reconstructed. Our NHSN archive contains historical counts, but report versions start on 19 November 2024. Our 2023/24 forecasts currently use latest

revised counts for past target weeks eligible at the generation date. Feedback through target week 27 April 2024 also uses latest counts, with an assumed release the following Wednesday (`synthetic_final`); this cannot assess the benefit of waiting for revisions. Public historical HHS reports contain earlier versions for both diseases. Delphi aggregates these to weekly counts for its 2023/24 base-forecast replay, but uses that season for initialization rather than a full replay of calibration feedback during 2023/24. We have not yet incorporated this reconstruction; weekly alignment and differences from NHSN require validation. From 20 November 2024, our base forecasts use reports available at their generation dates. The active hub design does not use synthetic earlier feedback.

Other limitations of the longer-history experiment. These are not flu-specific. The generated COVID base has zero width at horizon -1: its target is already a predictor, so all quantiles coincide before recalibration. The default regression solver could stall in this case; using the production solver `fn` resolved that problem. Target weeks 4 May–2 November 2024 have incomplete reporting and are excluded from learning and scoring; our forecast schedule pauses across this gap. Stateful methods carry accumulated corrections and weights forward when rounds resume, rather than restarting. The missing report versions and this reporting gap also affect COVID, despite its available training inputs.

Cleaning and filling.

1. **Missing forecasts.** Fill gaps of at most six rounds from the most recent earlier forecast for the same target: horizon h at reference date r uses $h + k$ at $r - 7k$ days, with the smallest available $k > 0$. Horizon 3 has no source. Filled rows are flagged and scored; unfillable rounds remain absent.
2. **Broken submissions.** Remove the known Delphi bugs: COVID 23 November 2024; COVID $h = -1$ on 25 April 2026; US flu 14 and 21 December 2024 and 4 January 2025. Also remove all-zero submissions across every location and quantile of a disease–round–horizon (flu $h = -1$, 22 November 2025).
3. **Missing reports.** If unpublished or currently invalid, stateful feedback waits and re-runs omit it; neither substitutes an older valid report. Keep the forecast timeline when latest truth is unavailable, but omit that target from scoring; earlier valid feedback can still be used. Targets 4 May–2 November 2024 are excluded from learning and scoring.
4. **Report errors.** A reviewed exclusion table masks identified erroneous report versions, retaining later corrections and their target weeks. Each exclusion records its evidence; decisions based on later revisions are retrospective cleaning. The existing performance results predate this filter. For example, Pennsylvania COVID for 16 November 2024 changed from 6,124 to 1,005 the next day.

Disease	Missing reference weeks	Unfilled target weeks, $h = 2$
COVID	25 January–1 February 2025	15 February 2025
Both	25 October–15 November 2025	15–29 November 2025
Flu	25 January 2025; 18 April 2026	None
Flu	7 June–11 October 2025	21 June–25 October 2025
Flu	6 June–3 October 2026	20 June–17 October 2026

The COVID $h = 2$ weight plot (`multiqt_weight_timelines/covid_horizon2_delay0.pdf`, p. 1) shows the first two gaps; only the first missing round can use an earlier $h = 3$ forecast. US flu also lacks target 4 January 2025 at $h = 2$ after successive broken submissions. Lighter bars denote filled rows. Lines break at gaps longer than 28 days and can bridge shorter missing intervals. Updates occur only at retained rounds: pending feedback waits across unfilled rounds, without resetting the state.

Locations. Every disease–location–horizon case learns independently. Detailed plots show the US, California, Texas, New York and Pennsylvania; this is the selected list, not the four largest states by population. Scores cover all 53 locations. Tuning and pooled summaries use the 52 non-US locations; the US is evaluated separately with their selected parameters. Forecasts are never averaged across locations.

Report screening. Large revisions and unusual weekly changes flag reports for review; they do not establish an error. Review the dated reports and their corrections before excluding a version.

3.2 Feedback

Feedback uses the latest report at each cutoff as defined in Section 3.1, independently of the recalibration method. Retrospective initialization is described in Section 3.3.

Stateful learning. A valid report is finite, nonnegative and not excluded by review. Let ν_s be the publication date of the first such report for row s. Wait d calendar weeks, giving maturity date $m_s^{(d)}$, then consume the newest report when valid at the next retained forecast cutoff with $t > s$. Write $\tau_d(s)$ for that round and $\mathcal{R}_t^{(d)}$ for the rows consumed before forecasting at t:

$$m_s^{(d)} = \nu_s + 7d \text{ days}, \tag{27}$$

$$\tau_d(s) = \min\{t > s : \iota_t \geq m_s^{(d)},\ y_s^{[\iota_t]} \text{ valid}\}, \qquad \mathcal{R}_t^{(d)} = \{s : \tau_d(s) = t\}. \tag{28}$$

An invalid newest report postpones learning; an older version is never substituted. Use the report at actual consumption time ι_t, even if a missing forecast delayed it beyond maturity. Each row is learned at most once; subsequent revisions are ignored. The report used for a consumed row is

$$\bar{y}_s^{(d)} = y_s^{[\iota_{\tau_d(s)}]}, \qquad \tau_d(s) \text{ exists}. \tag{29}$$

Plain MultiQT on counts starts at $\tilde{\theta}_0 = 0$ and, once active, updates against previously returned forecasts before computing q_t:

$$\tilde{\theta}_t^{\alpha} = \tilde{\theta}_{t-1}^{\alpha} - \eta_t \sum_{s \in \mathcal{R}_t^{(d)}} (\mathbf{1}\{y_s^{[\iota_t]} \leq q_s^{\alpha}\} - \alpha), \tag{30}$$

$$\theta_t = \Pi_{\mathcal{K}-b_t}(\tilde{\theta}_t), \qquad q_t = b_t + \theta_t. \tag{31}$$

The hidden state is unprojected; an empty release set gives no update. The `log1p` tracker applies the same rule in ψ coordinates.

Weekly re-run. For forecasts and states, a superscript $[t]$ identifies the re-run at reference week t: $q_s^{[t]}$ is its reconstructed forecast for row s. This differs from the report cutoff date v in $y_s^{[v]}$. At reference week t, reconstruct plain MultiQT from zero using reports current at ι_t, retaining the release positions $j = \tau_d(s)$ and delay d, but omitting invalid current reports:

$$\tilde{\theta}_j^{\alpha,[t]} = \tilde{\theta}_{j-1}^{\alpha,[t]} - \eta_j^{[t]} \sum_{\substack{s \in \mathcal{R}_j^{(d)} \\ y_s^{[\iota_t]} \text{ valid}}} (\mathbf{1}\{y_s^{[\iota_t]} \leq q_s^{\alpha,[t]}\} - \alpha), \tag{32}$$

$$\theta_j^{[t]} = \Pi_{\mathcal{K}-b_j}(\tilde{\theta}_j^{[t]}), \qquad q_j^{[t]} = b_j + \theta_j^{[t]}. \tag{33}$$

Only $q_t^{[t]}$ is returned; earlier returned forecasts are unchanged. All re-runs keep the stateful S_0 and L_0 fixed (Section 3.3). Studentized methods use (22), reconstruct their scale history and return the base until a required scale exists. An invalid revised report removes an update, not an expert: the declared score bank remains fixed (Table 3).

Report filtering. Select the latest report at the cutoff, then apply the reviewed exclusions; never substitute an older or future report. Count the delay from the first acceptable report. Stateful updates wait for an acceptable current report; re-runs omit excluded versions. Recompute S_0, L_0 from the filtered stateful schedule, then keep them fixed across re-runs. Use the latest version for scoring and peak detection if acceptable; otherwise leave that target unscored for every method. Apply the same exclusions to the training reports of our own base forecasts.

3.3 Tuning

Reference scales use the first 20 consumed rows in release order, $I_{20}^{(d)}$, which need not span 20 calendar weeks. Parameter selection uses the tuning period; these are separate operations.

Reference scales and retrospective initialization. Write Q_p for an empirical p-quantile. The initial error scale S_0, distinct from the score function S_t, pools absolute base errors in tracker units:

$$S_0 = \max\Big\{\varepsilon_S, Q_{0.9}\Big(\Big\{|\psi(b_s^\alpha) - \psi(\bar{y}_s^{(d)})| : s \in I_{20}^{(d)},\ \alpha \in \mathcal{A}\Big\}\Big)\Big\}, \qquad \varepsilon_S = 10^{-6}. \tag{34}$$

It sets rates $\eta = \kappa S_0$, with rate multiplier κ, and studentized scale bounds $[.05S_0, S_0]$; studentized step sizes are selected in standardized units. Script 02 labels the multiplier m, distinct from $m = |\mathcal{A}|$ here. The reference loss L_0 is defined in Section 3.5.3.4.

With the default `replay_warmup=TRUE`, both constants use the eventual first 20 releases and apply from the start. Forecasts before the 20th release therefore use retrospective scale information. In the active data these rows have tuning target dates, but some reports arrive after early evaluation cutoffs: tuning-period membership does not imply availability before evaluation. With replay disabled, learning waits for warmup; fewer than 20 releases leaves learning inactive. The constants stay fixed across weekly re-runs; states, weights and time-varying scales are reconstructed. Evaluation never resets the tracker, so its initial evaluated correction is generally nonzero.

Cases and selection. Tune COVID before flu, and `log1p` before `identity` within each disease. For each pair, run stateful delay 0, delay 1 and weekly re-run at delay 0. These 12 configurations have two stages each (24 total). Each starts from defaults; configurations without tuning rows stop. With hub forecasts, tune horizons $-1, \ldots, 3$. With our own base forecasts, tune $0, \ldots, 3$, with -1 reusing their choices because the base has zero width there.

Candidates replay the full history with common S_0, L_0 within each case, tracker and feedback design. Only tuning target weeks are scored, including initialization rows when valid. Scores use latest archived truth, which need not have been available before evaluation. Pool case-level means equally over non-US locations and horizons. The criterion is log quantile loss for `log1p` and quantile loss per 100,000 for `identity`, without L_0 normalization.

Stage 1 selects tracker rates separately for `fixed`, `leaky`, `studentized`, `studentized_base`, `median_constant`, `studentized_full` and `studentized_level_rms`. Let $\overline{\mathcal{L}}_j(\vartheta)$ be the mean loss of method j at candidate ϑ over tuning cases, and let $\overline{\mathrm{CE}}_j(\vartheta)$ be its mean calibration error. Subscript 0 denotes the original forecast. Stage 2 holds the selected rates fixed and chooses H separately for `weighted`, `weighted_common`, `shared`, `shared_common`, `restart_weighted` and `score_learned`. Both stages minimize the sum of percentage changes from the original:

$$E_j(\vartheta) = 100\left[\frac{\overline{\mathcal{L}}_j(\vartheta)}{\overline{\mathcal{L}}_0} - 1\right] + 100\left[\frac{\overline{\mathrm{CE}}_j(\vartheta)}{\overline{\mathrm{CE}}_0} - 1\right]. \tag{35}$$

The code calls this the elbow rule: one percentage point of loss change and one percentage point of calibration-error change have equal weight. It does not estimate curvature. Ties select the

smaller parameter value. This single pass avoids a Cartesian search; earlier rate choices are not revisited and need not give the best joint combination.

Stage	Setting	Candidates
1	`fixed_multiplier`	.001, .0025, .005, .01, .02, .05, .1, .2, .4
1	`median_rate`	.003, .01, .03, .1, .3, 1
1	`leaky_multiplier`	.001, .002, .004, .008, .02, .05, .1, .2
2	`H`	2, 4, 8, 16, 32

Multipliers scale S_0. `median_rate` is standardized. H counts processed weight updates and also sets sharing $\gamma = 1/(2H)$ where enabled.

Fixed and inherited settings. Learned mixtures use adaptive λ from $\lambda_{\text{start}} = 2.72$; $\lambda = 0$ gives equal weights in `exp_weights()`. Recent-start count is 8; outer and Decline rates are $.001S_0$ and $.05S_0$; the declining-context threshold is 0. Other constants remain as specified with each method. `studentized_zero` and `median_constant_zero` inherit their corresponding normal-start rates, and outer variants retain the selected settings of their inner mixtures.

Tuning plots. The current `multiqt_tuning.pdf` pairs calibration error with quantile loss, both as percentage changes from the original. Solid/dashed lines use the log/identity criteria. Symbol size grows with the candidate value; circles mark selected values and an open square at $(0, 0)$ marks the original. On pooled pages, a dotted line of slope -1 through each selected point shows its value of (35); no candidate from that curve lies below it. Pooled pages by disease and feedback design precede pages for California, Texas, New York and Pennsylvania, when available. State pages average over horizons but circle the pooled selection. H curves condition on selected rates. The `on_grid_edge` flag marks an endpoint selection, not evidence of an interior optimum.

3.4 Evaluation

Peaks and evaluation windows. Let $y_t^\star$ be the latest archived count for the target of row t. For peak detection, script 01 omits invalid counts, including reviewed exclusions, and the incomplete-reporting interval. `find_peaks()` then uses three steps:

1. Smooth with a centred three-observation mean, leaving endpoints unchanged. A candidate is the maximum within eight retained observations on either side; ties go to the earliest.
2. Require positive *prominence*: the smoothed height above the higher of the two surrounding minima, searching on each side until a strictly higher value or the series end. It must reach at least 30% of the candidate height and 10% of the largest retained raw count on a forecast target date.
3. Move the peak to the largest raw count within one retained observation, earliest at ties; keep dates within the forecast target range.

These neighbourhoods count observations, not calendar weeks across gaps. Peaks are hindsight labels, used for prediction only by the target oracle.

Let T_t be the target date, $p(t)$ its nearest retained peak date (earlier at ties), and $(T_t - p(t))/7$ their distance in calendar weeks. For evaluation rows E with valid latest truth, score the windows

$$\begin{aligned} W_{\text{overall}} &= E, \\ W_{\text{before}} &= \{t \in E : -7 \le (T_t - p(t))/7 \le 0\}, \\ W_{\text{after}} &= \{t \in E : 1 \le (T_t - p(t))/7 \le 8\}. \end{aligned} \tag{36}$$

Rows count once; the peak week belongs to "up to peak". Overall also includes weeks outside these windows. Missing rows and period boundaries shorten windows without discarding peaks. Period membership follows the target date; its nearest peak may lie in another period. The main plots use these windows over the full evaluation period.

Loss and calibration error. A case fixes location, horizon and feedback design within a disease. For a selected window W of $N = |W|$ rows, average over rows and all 23 quantile levels:

$$\mathcal{L}^{\mathrm{nat}}(W) = \frac{1}{mN} \sum_{t \in W} \sum_{\alpha \in \mathcal{A}} \rho_\alpha(q_t^\alpha / c_{\mathrm{scale}}, y_t^\star / c_{\mathrm{scale}}), \tag{37}$$

$$\mathcal{L}^{\log}(W) = \frac{1}{mN} \sum_{t \in W} \sum_{\alpha \in \mathcal{A}} \rho_\alpha(\log(1 + \max\{q_t^\alpha, 0\}), \log(1 + y_t^\star)), \tag{38}$$

$$\mathrm{CE}(W) = \frac{1}{m} \sum_{\alpha \in \mathcal{A}} \left| \frac{1}{N} \sum_{t \in W} \mathbf{1}\{y_t^\star \le q_t^\alpha\} - \alpha \right|. \tag{39}$$

Columns are `quantile_loss`, `log_quantile_loss` and `coverage_error`; tuning uses `loss`, `log_loss` and `coverage_error`. CE averages weeks before taking the absolute residual. All feedback designs use the same latest truth; empty windows give missing scores. Evaluation losses are never divided by L_0. Scripts 04/06 display the criterion of the tracker; script 02 displays loss per 100,000.

Quantile-level diagnostics. `multiqt_by_level/delay0.pdf` repeats the fixed-rate trade-off at each level:

$$\mathcal{L}_\alpha^{\mathrm{nat}}(W) = \frac{1}{N} \sum_{t \in W} \rho_\alpha(q_t^\alpha / c_{\mathrm{scale}}, y_t^\star / c_{\mathrm{scale}}),$$
$$\mathrm{CE}_\alpha(W) = \left| \frac{1}{N} \sum_{t \in W} \mathbf{1}\{y_t^\star \le q_t^\alpha\} - \alpha \right|. \tag{40}$$

Pages fix disease and level, with location rows, horizon columns and phase symbols. There are 23 COVID pages then 23 flu pages, in increasing α order; levels .1, .5, .9 appear on pages 4, 12, 20 (COVID) and 27, 35, 43 (flu).

All-location trade-offs. Write $\mathcal{L}_{j,c}$ and $\mathrm{CE}_{j,c}$ for the mean quantile loss and calibration error of method j in case c; $j = 0$ is the original. A set $\mathcal{C}$ contains J cases. In `multiqt_peak_reasoning_tradeoff.pdf`, columns are horizons $-1, \ldots, 3$; blocks are delay 0, delay 1 and re-run, each showing overall, up to peak and after peak. Lower left is better. The first page per disease shows individual non-US cases and large method means

$$\left(J^{-1} \sum_c \mathcal{L}_{j,c},\ J^{-1} \sum_c \mathrm{CE}_{j,c} \right).$$

These average location scores, not forecasts or pooled coverage indicators. The first row of `multiqt_error_tradeoff.pdf` uses the same aggregation for fixed rates. Feedback designs have separately selected parameters, so their differences include retuning. Axes vary by panel; all-location limits use the 1st–99th percentiles and mean points, with loss capped near four times the base. Extremes can be hidden or marked `>`. Original/common-weight/outer methods use open circles/triangles/squares or diamonds; exact forecast duplicates are hidden.

Evaluation tree. Script 06 splits pooled disease cases by feedback design, then horizon. For each window it reports

$$R_j(\mathcal{C}) = 100\left[1 - \frac{\sum_{c\in\mathcal{C}} N_c \mathcal{L}_{j,c}}{\sum_{c\in\mathcal{C}} N_c \mathcal{L}_{0,c}}\right], \tag{41}$$

$$P_j(\mathcal{C}) = J^{-1}\sum_{c\in\mathcal{C}} \mathbf{1}\{\mathcal{L}_{j,c} < \mathcal{L}_{0,c}\}. \tag{42}$$

`reduction` is a ratio of week-weighted losses (positive is better); `share_improved` weights cases equally, counts ties as not improved, and is printed for at least five cases. CE panels also average case scores with week weights. Each page displays the ten methods with lowest pooled after-peak loss: a descriptive evaluation ranking, not tuning. Overlapping horizons/designs are not independent replicates. Tree panels share axes; clipping is marked `<` or `>`. By contrast, script 02 phase-rate plots show loss change $100(\mathcal{L}_j/\mathcal{L}_0 - 1)$, where negative is better.

Forecast curves. Files in `multiqt_method_forecasts/` show levels $.1, \ldots, .9$ and weekly loss per 100,000 at a fixed horizon; gray marks tuning, dotted lines peaks. Files in `multiqt_season_forecasts/` add coverage and loss trajectories for selected fixed rates. `multiqt_base_forecast_paths.pdf` shows full count timelines, median paths around all retained peaks and three diagnostic indicators (Section 4.3). It spans tuning and evaluation; its flu winter 2024/25 peak is within tuning, not held-out evidence. Weight timelines show how each mixture combines its experts. Gap dates are listed in Section 3.1.

3.5 Methods

The methods below use the preceding data, feedback, tuning and evaluation conventions unless stated otherwise.

3.5.1 Single trackers

These methods maintain one MultiQT tracker for each case.

Fixed rate: `fixed`. Use (30)–(31) with $\eta = \kappa S_0$ in tracker units. Tune κ separately (default .05). Corrections persist across rounds.

Leaky tracker: `leaky`. Add a common decaying shift to persistent MultiQT offsets, as in (61)–(62). The separately tuned method also belongs to the score bank.

Persistent restart control: `restart_persistent`. Keep the first eligible weekly-start expert with $\kappa = .05$ (Section 3.5.3.5). It learns only from forecasts made since its start. Its rate is not separately tuned, so this control for restart mixtures need not reproduce `fixed`.

3.5.2 Studentized scores

Studentized variants and median initialization. In tracker units, use centre $a_t^\alpha = \beta_t^\alpha$ for the full base or $a_t^\alpha = \beta_t^{.5}$ for its median. After processing available feedback, return

$$q_t = \psi^{-1}\left[\Pi_{\mathcal{K}}^{A_t^{-2}}(a_t + A_t\tilde{u}_t)\right]. \tag{43}$$

For a common scale, the projection is the ordinary isotonic projection. After warmup, the delayed update is applied to the hidden state,

$$\tilde{u}_t^\alpha = \tilde{u}_{t-1}^\alpha + \eta \sum_{s\in\mathcal{R}_t^{(d)}} \left(\alpha - \mathbf{1}\{\bar{y}_s^{(d)} \le q_s^\alpha\}\right). \tag{44}$$

Compare each report with its historical forecast q_s, reconstructed inside a re-run when applicable. Project only the forecast; the hidden state carries forward throughout the run. The standardized step has default $\eta = .1$ and is tuned for each normal-start or full-base method. Zero-start variants inherit the corresponding normal-start rate.

Scales from recent errors or the current forecast. Let $\mathcal{H}_t^{(d)}$ contain the last five consumed rows, or fewer if unavailable, and set $e_s^\alpha = \psi(\bar{y}_s^{(d)}) - \beta_s^\alpha$. These need not be consecutive calendar weeks. A hat marks a raw scale estimate, and Φ is the standard normal distribution function. The default scales are

$$\widehat{\sigma}_{t,\text{med}} = \left[\frac{1}{|\mathcal{H}_t^{(d)}|}\sum_{s\in\mathcal{H}_t^{(d)}}(e_s^{.5})^2\right]^{1/2}, \tag{45}$$

$$\widehat{\sigma}_{t,\text{res}}^\alpha = \left[\frac{1}{|\mathcal{H}_t^{(d)}|}\sum_{s\in\mathcal{H}_t^{(d)}}(e_s^\alpha)^2\right]^{1/2}. \tag{46}$$

Both require five consumed rows. RMS includes bias because the errors are not centred. The optional robust alternative is median $|e_s^\alpha|/\Phi^{-1}(.75)$.

The common scale inferred from the current base is

$$\widehat{\sigma}_{t,\text{base}} = \frac{\beta_t^{.9} - \beta_t^{.1}}{\Phi^{-1}(.9) - \Phi^{-1}(.1)}. \tag{47}$$

The denominator is approximately 2.563. Base-width scales respond immediately to changes in the current forecast; residual scales respond only after the corresponding reports arrive.

Floor raw scales at 10^{-6}, then clip to $[.05S_0, S_0]$ once S_0 exists; denote these bounded scales by σ. The bounds limit how much the scale suppresses or amplifies accumulated corrections. They are implementation defaults, not requirements of studentization.

Table 1: Studentized methods. Scales refer to the bounded versions above. Initial states are specified before the first update.

Method ID	Centre and scale	Initialization and purpose
`studentized`	$a_t^\alpha = \beta_t^{.5}$; common $\sigma_{t,\text{med}}$	$\tilde{u}_{\text{init}}^\alpha = \Phi^{-1}(\alpha)$. Reconstruct the distribution around the base median using recent errors.
`studentized_base`	$a_t^\alpha = \beta_t^{.5}$; common $\sigma_{t,\text{base}}$	Normal initial state. Reconstruct the distribution from the median and the central width of the current base forecast.
`median_constant`	$a_t^\alpha = \beta_t^{.5}$; common constant σ	Normal initial state; freeze $\sigma_{t,\text{med}}$ at the first reference week after both required warmups. Separates replacing the base by its median from subsequently changing the scale.
`studentized_zero`	$a_t^\alpha = \beta_t^{.5}$; common $\sigma_{t,\text{med}}$	$\tilde{u}_{\text{init}} = 0$. Start all levels at the median; use the rate selected for `studentized`.
`median_constant_zero`	$a_t^\alpha = \beta_t^{.5}$; common constant σ	Zero initial state; use the scale and rate of `median_constant`.

Continued on next page

Table 1 (continued)

Method ID	Centre and scale	Initialization and purpose
`studentized_full`	$a_t^\alpha = \beta_t^\alpha$; common $\sigma_{t,\text{base}}$	$\tilde{u}_{\text{init}} = 0$. Keep the full base and rescale its accumulated corrections together.
`studentized_level_rms`	$a_t^\alpha = \beta_t^\alpha$; $\sigma_{t,\text{res}}^\alpha$	Zero initial state. Set the scale at each level from recent errors at that level; weighted isotonic projection.

The scale of `median_constant` is fixed within each reconstructed history but can change between re-runs; S_0 and L_0 cannot. Zero-start variants compare initialization at the same selected rate. Their first active forecast may already include a feedback update.

The residual-median methods also allow $t_5^{-1}(\alpha)$ initialization; the base-width median method always uses normal quantiles. It retains the median and bounded central width but imposes a symmetric normal shape in tracker units, so it need not reproduce the original forecast. Until a required residual scale exists, return the original full base. Feedback consumed while waiting is not replayed; later feedback on those rounds is compared with the original forecasts.

Studentization rescales the accumulated correction, rather than only the newest increment. For full-base `log1p` methods without crossings,

$$q_t^\alpha = (1 + b_t^\alpha)\exp(\sigma_t^\alpha \tilde{u}_t^\alpha) - 1, \tag{48}$$

so the count correction is non-additive. With `identity`, it is additive with a changing multiplier. Quantile-specific scales retain the coverage target, but no calibration guarantee is established here for those variants.

3.5.3 Weighted experts

An expert is a complete forecasting procedure; a bank is the set being combined. Weights learn from quantile losses separately for each location–horizon case. A fixed learning rate still changes the weights; an adaptive rate changes their responsiveness. The formulas first assume a fixed bank with all experts eligible.

3.5.3.1 Mixtures and feedback losses For K experts, let $f_{k,t}^\alpha$ be expert k's forecast and w_k^α its weight. Suppressing time indices for one forecast, form the mixture $\hat{q}$ and repair any crossings:

$$\hat{q}^\alpha = \sum_{k=1}^K w_k^\alpha f_k^\alpha, \qquad w_k^\alpha \geq 0, \quad \sum_{k=1}^K w_k^\alpha = 1, \qquad q = \Pi_{\mathcal{K}}(\hat{q}). \tag{49}$$

Per-quantile weights can introduce crossings, repaired by isotonic projection. Common weights replace w_k^α by w_k and preserve order when every expert is ordered. All mixtures average counts, including for `log1p` trackers.

Let n count processed feedback updates, with forecast row $s(n)$ and report Y_n. Score the expert forecasts for that row, using L for unnormalized losses and ℓ for losses normalized by L_0:

$$L_{k,n}^\alpha = \rho_\alpha\left(f_{k,s(n)}^\alpha / c_{\text{scale}}, Y_n / c_{\text{scale}}\right), \qquad L_{k,n} = \frac{1}{m}\sum_{\alpha\in\mathcal{A}} L_{k,n}^\alpha, \tag{50}$$

$$\ell_{k,n}^\alpha = L_{k,n}^\alpha / L_0, \qquad \ell_{k,n} = \frac{1}{m}\sum_{\alpha\in\mathcal{A}} \ell_{k,n}^\alpha. \tag{51}$$

The same L_0 applies to all experts and levels within a case and feedback design (Section 3.5.3.4). Score the originally computed forecasts in stateful learning and the reconstructed forecasts in a re-run.

3.5.3.2 Exponential updates and optional sharing Let $w_{k,n}^{\alpha}$ and λ_n^{α} be the weight and rate before update n. Before optional sharing, update separately at each level to obtain $\bar{w}_{k,n+1}^{\alpha}$:

$$\bar{w}_{k,n+1}^{\alpha} = \frac{w_{k,n}^{\alpha} \exp(-\lambda_n^{\alpha} \ell_{k,n}^{\alpha})}{\sum_{j=1}^{K} w_{j,n}^{\alpha} \exp(-\lambda_n^{\alpha} \ell_{j,n}^{\alpha})}. \tag{52}$$

With common weights, there is one update using the mean loss:

$$\bar{w}_{k,n+1} = \frac{w_{k,n} \exp(-\lambda_n \ell_{k,n})}{\sum_{j=1}^{K} w_{j,n} \exp(-\lambda_n \ell_{j,n})}. \tag{53}$$

Lower loss increases relative weight; larger λ strengthens the response. Weights start uniformly, and $\lambda = 0$ keeps equal weights.

When sharing is enabled with fraction γ, follow the exponential step by

$$w_{k,n+1}^{\alpha} = (1-\gamma)\bar{w}_{k,n+1}^{\alpha} + \gamma/K. \tag{54}$$

Omit α for common weights. Sharing helps downweighted experts recover: use $\gamma = 1/(2H)$ when enabled and zero otherwise. Table 2 distinguishes sharing from common weights across levels.

3.5.3.3 Adaptive learning rates The lookback H counts processed updates. The first H updates use starting rate λ_{start}. For rate-grid and weekly-start mixtures, record the weighted second moment of the losses using weights before the exponential step and sharing:

$$\begin{aligned} M_n^{\alpha} &= \sum_{k=1}^{K} w_{k,n}^{\alpha} (\ell_{k,n}^{\alpha})^2, \\ M_n &= \sum_{k=1}^{K} w_{k,n} \ell_{k,n}^2. \end{aligned} \tag{55}$$

The score mixture `score_learned` instead uses the weighted variance (`centered=TRUE`):

$$M_n^{\alpha} = \sum_{k=1}^{K} w_{k,n}^{\alpha} \left(\ell_{k,n}^{\alpha} - \sum_{j=1}^{K} w_{j,n}^{\alpha} \ell_{j,n}^{\alpha} \right)^2. \tag{56}$$

For this centred rule, treat variances at most $10^{-12} \sum_k w_{k,n}^{\alpha} (\ell_{k,n}^{\alpha})^2$ as zero, and analogously for common weights, to avoid large rates from numerical rounding. For per-quantile weighting, the rate is

$$\lambda_n^{\alpha} = \begin{cases} \lambda_{\text{start}}, & 1 \le n \le H, \\ \sqrt{\dfrac{\log(2KH)+1}{\sum_{r=n-H}^{n-1} M_r^{\alpha}}}, & n > H. \end{cases} \tag{57}$$

Only the previous H moments enter; a zero sum retains the previous finite rate. Larger moments reduce the rate. For the centred score mixture, a loss shared by all experts does not increase the variance.

Per-quantile weighting maintains a vector, moment history and rate for each level. Common weighting uses one of each: average losses across levels *before* computing M_r, then omit α in (57). This is not an average of separately learned weights or per-quantile moments.

Later starts and implementation. Only restart experts that forecast the released row are eligible. Normalize their weights for the update, then restore their total mass; other weights stay unchanged. Use the eligible count in the rate formula. A new expert receives $1/K$ of the enlarged bank, preserving existing relative weights. Updates with fewer than two eligible experts do not count toward H.

3.5.3.4 Expert loss normalization Using the quantile loss in (1), define one reference loss per case and delay:

$$L_0 = \max\left\{\varepsilon_L, \frac{1}{20|\mathcal{A}|}\sum_{s\in I_{20}^{(d)}}\sum_{\alpha\in\mathcal{A}}\rho_\alpha\left(\frac{b_s^\alpha}{c_{\text{scale}}}, \frac{\bar{y}_s^{(d)}}{c_{\text{scale}}}\right)\right\}, \qquad \varepsilon_L = 10^{-6}. \tag{58}$$

Losses are relative to the initial mean base loss: $\ell = .8$ means 80% of that reference, not necessarily a 20% improvement in the current week. Values can exceed one. Weighting uses scaled counts even for `log1p` trackers.

Initial and fixed-rate weight updates. Dividing by L_0 preserves rankings and makes weight changes depend on relative loss gaps. A change of count units scales both L and L_0 equally, leaving weights unchanged when the numerical floor is inactive. This matters throughout fixed-rate runs and during the first H adaptive-run updates at λ_{start}; these are distinct from the 20 rows used to estimate L_0.

For equal initial weights, losses 80 and 100, $L_0 = 100$ and $\lambda_{\text{start}} = 1$ give weights about .55 and .45 before sharing. The same numerical rate on raw losses would put almost all weight on the first expert.

Effect on adaptive learning rates. After H updates, the rule in (57) already adapts to the loss moments. Use superscripts raw, norm for calculations without and with L_0 normalization. Conditional on the same pre-update weight vectors at updates $n-H,\dots,n$ and a positive moment sum, normalization gives, separately at each level,

$$\begin{aligned} M_r^{\alpha,\text{norm}} &= M_r^{\alpha,\text{raw}}/L_0^2, & \lambda_n^{\alpha,\text{norm}} &= L_0\lambda_n^{\alpha,\text{raw}}, \\ \lambda_n^{\alpha,\text{norm}}\ell_{k,n}^\alpha &= \lambda_n^{\alpha,\text{raw}}L_{k,n}^\alpha. \end{aligned} \tag{59}$$

The identity holds for both moment definitions and for common weights. The centred rule's numerical zero-variance threshold scales by L_0^{-2} too. Thus the loss and rate changes cancel in the exponential update, conditional on the same weight history. Runs with the same numerical starting rate can still differ: their first H updates change later weights and moments.

Equivalent runs require matching settings and $\lambda_{\text{start}}^{\text{raw}} = \lambda_{\text{start}}^{\text{norm}}/L_0$, including the same scaling of retained rates when the moment sum is zero. Keep normalization throughout a run so incoming losses, stored moments and rates remain in consistent units. After adaptation, L_0 is not an additional reduction in update strength.

Effect across quantile levels and re-runs. A single L_0 does not equalize losses across levels. Per-quantile weights use the loss and adaptive rate of each level; common weights use mean losses. Re-runs reconstruct losses but retain the stateful L_0. Evaluation losses are not normalized by L_0, and normalization alone guarantees neither accuracy nor calibration.

3.5.3.5 Expert combinations implemented in the code Table 2 groups the current combinations. Every learned mixture uses the adaptive rate above, with second moments for rate-grid and weekly-start mixtures and variance for the score mixture.

Method IDs	Weighting and bank
`uniform`	Equal weights on 11 persistent rate experts.
`weighted`, `shared`	Same rate bank; per-quantile weights, without/with sharing.
`weighted_common`, `shared_common`	Same rate bank; common weights, without/with sharing.
`restart_uniform`	Equal weights on all weekly starts, rate $.05S_0$.
`restart_weighted`	Same starts; adaptive λ, sharing, per-quantile weights.
`restart_recent`	Equal weights on the eight most recent starts, rate $.05S_0$.
`score_equal`	Equal weights on the declared score bank.
`score_learned`	Same score bank; adaptive λ, sharing, per-quantile weights.

Table 2: Current expert combinations.

The four `_outer` variants of `shared`, `shared_common`, `restart_uniform`, and `restart_weighted` add persistent MultiQT at $.001S_0$ to their inner mixture, retaining its settings and learning from their own final forecasts. Rate and H selection follow Section 3.3.

Rate and weekly-start banks. The rate bank uses

$$\kappa \in \{0, .01, .015, .02, .03, .05, .075, .1, .15, .2, .3\};$$

all experts start together and the zero-rate member is the original. Weekly-start banks add a fresh copy at every retained reference week when initialization permits; with warmup replay, starts begin in the first round. Each copy starts with zero correction and learns only from feedback on forecasts made since that start. Older copies keep their accumulated corrections. There is no permanent zero-rate expert.

For `restart_recent`, $\mathcal{B}_t$ contains the most recent $K_{\text{recent}} = 8$ copies, or all copies if fewer have started. Then

$$q_t^\alpha = |\mathcal{B}_t|^{-1} \sum_{k \in \mathcal{B}_t} f_{k,t}^\alpha. \tag{60}$$

This controls the age of corrections without detecting peaks. Count starts, not elapsed weeks across gaps. A new expert receives weight before feedback, so a new color band alone does not show lower loss.

The six score-based experts and fixed membership. The bank compares ways to retain or rebuild the base and correct its errors. Write $S_{k,t}^\alpha(y)$ for member k's residual score. For the scores below, $r = \psi(y)$; project and invert as in (43). Fixed, leaky and studentized experts use separately selected rates; Decline uses $.05S_0$ and the original does not update. This compares complete procedures, not scores at a common rate.

Membership is fixed across reference weeks and re-runs, even when forecasts coincide. Exclude Decline at $h = -1$, where it equals the original by construction. Do not deduplicate complete forecast histories: later forecasts or revisions must not change earlier membership.

Table 3: The declared score bank. Scales are the bounded scales defined earlier; Decline is omitted at $h = -1$.

Member	Score or forecast	Purpose
`original`	$f_{k,t}^\alpha = b_t^\alpha$	Preserve the base as an option.

Member	Score or forecast	Purpose
`fixed`	$S^{\alpha}_{k,t}(y) = r - \beta^{\alpha}_{t}$	Persistent additive correction; $\eta = \kappa S_0$.
`leaky`	Fixed MultiQT plus v_t in (61)	Short-lived common directional correction alongside persistent offsets.
`studentized_base`	$S^{\alpha}_{k,t}(y) = (r - \beta^{5}_{t})/\sigma_{t,\text{base}}$	Rebuild around the median with current central width and normal initial thresholds.
`studentized_full`	$S^{\alpha}_{k,t}(y) = (r - \beta^{\alpha}_{t})/\sigma_{t,\text{base}}$	Keep the full base; rescale accumulated corrections with its central width.
`decline`	$S^{\alpha}_{k,t}(y) = r - \beta^{\alpha}_{t}$, updated only for $c^{\text{ctx}}_{s,h} = 1$	Learn and apply a persistent correction in declining forecast context; return the original otherwise.

Initialization and scales follow Table 1. Leaky and Decline change state updates, rather than defining new static residual scores. `studentized_level_rms` is a standalone method.

The leaky expert. Besides its persistent MultiQT offsets, this expert has one common shift v_t on the log scale, initially zero. Let Δ_t be elapsed calendar weeks since the previous reference date, at least one. Write $\text{clip}_{[a,b]}(x) = \min\{b, \max\{a, x\}\}$. After warmup, its update is

$$v_t = \text{clip}_{[-.2,.2]}\left[2^{-\Delta_t} v_{t-1} + .2 \sum_{s \in \mathcal{R}^{(d)}_t} \frac{1}{m} \sum_{\alpha \in \mathcal{A}} \left(\alpha - \mathbf{1}\{\bar{y}^{(d)}_s \le f^{\alpha}_{\text{leaky},s}\}\right)\right], \tag{61}$$

The shift halves per calendar week, including gaps. Sum reports within a batch, then cap the shift. Errors in the same direction reinforce it; opposite errors across levels can cancel. It therefore targets a common directional error. Ordinary offsets persist and update at each level.

For `log1p`, add v_t after projection and before inversion. For `identity`, apply it to the ordinary projected count forecast $q^{\alpha}_{\text{plain},t}$ as

$$f^{\alpha}_{\text{leaky},t} = (1 + (q^{\alpha}_{\text{plain},t})_+)e^{v_t} - 1 + \min(q^{\alpha}_{\text{plain},t}, 0). \tag{62}$$

This gives the same log-shift for nonnegative counts and extends it to negative forecasts. These trackers do not clip final forecasts at zero.

Target-specific signal and Decline. Let $Z(t, h; v)$ be $\log(1 + \text{base median})$ at reference date t and horizon h, if the median is finite, nonnegative, available by v and not filled; otherwise it is missing. Write $Z_{t,h} = Z(t, h; \iota_t)$. The signal compares the target horizon with the largest available earlier median in the same forecast path. It defines context $c^{\text{ctx}}_{t,h} \in \{0, 1\}$:

$$\begin{aligned} D_{t,h} &= \max_{\substack{r=-1,\ldots,h-1 \\ Z_{t,r} \text{ available}}} Z_{t,r} - Z_{t,h}, \\ c^{\text{ctx}}_{t,h} &= \mathbf{1}\{D_{t,h} > 0\}. \end{aligned} \tag{63}$$

The target median and at least one earlier median must be available by the cutoff at the exact reference date, excluding filled forecasts. Without this information, including at $h = -1$, set the context to zero. The threshold is fixed at zero: the target median must be strictly lower than

an earlier median in the same forecast path. The signal describes a decline in the base forecast path; it is not a calibrated probability of being after the true peak.

Decline starts with zero hidden offsets and uses

$$\tilde{\theta}_t^\alpha = \tilde{\theta}_{t-1}^\alpha + .05 S_0 \sum_{s \in \mathcal{R}_t^{(d)}} c_{s,h}^{\text{ctx}} \left(\alpha - \mathbf{1}\{ \bar{y}_s^{(d)} \le f_{\text{decline},s}^\alpha \} \right), \tag{64}$$

$$f_{\text{decline},t} = \begin{cases} \psi^{-1}\left[\Pi_{\mathcal{K}}(\beta_t + \tilde{\theta}_t) \right], & c_{t,h}^{\text{ctx}} = 1, \\ b_t, & c_{t,h}^{\text{ctx}} = 0. \end{cases} \tag{65}$$

The historical context of row s determines whether its feedback updates this state; the current context determines whether to apply the correction. Offsets persist between declining periods. The same warmup and re-run conventions apply as for Fixed MultiQT.

Equal and learned score mixtures. `score_equal` averages the declared experts. For `score_learned`, learn one vector per quantile level using the normalized losses, adaptive rate and sharing above, then repair crossings by isotonic projection.

3.5.4 Reference benchmarks

The references are the original forecast, a hindsight control and two production configurations of Delphi [3].

Original forecast: `original`. Return $q_t = b_t$ without recalibration or learning. This is the reference for the reported loss changes and reductions.

Target oracle: `oracle_target`. This hindsight control returns

$$q_t^{\text{oracle}} = \begin{cases} q_t^{\text{fixed}}, & T_t \le p(t), \\ b_t, & T_t > p(t). \end{cases} \tag{66}$$

This bypasses correction for postpeak targets, including forecasts made before the peak. Fixed MultiQT still learns from its own forecasts. The control tests bypassing correction with perfect phase information; it is not deployable. Setting the rate to zero would instead freeze the offset, giving a different control.

Delphi benchmarks: separate scales and initialization. Both run for both diseases and all feedback designs, independently of the identity/log setting. States start at zero without S_0 or warmup; constants are untuned. For transform χ and projected forecast ξ_t in those units, the released coverage-error sum is

$$\epsilon_t^\alpha = \sum_{s \in \mathcal{R}_t^{(d)}} \left(\alpha - \mathbf{1}\{ \chi(\bar{y}_s^{(d)}) \le \xi_s^\alpha \} \right). \tag{67}$$

In a weekly re-run, use its current reports and reconstructed forecasts, as in Section 3.2.

Adaptive fast and slow offsets: `ref_op`. Let $\mathcal{F}_t^{(d)} = \bigcup_{j \le t} \mathcal{R}_j^{(d)}$ contain all consumed rows, and $\mathcal{F}_{t,20}^{(d)}$ the last 20 in release order, or fewer if unavailable. The COVID production configuration uses $\chi(x) = \sqrt{x}$ on raw counts, with rates

$$\begin{aligned} \eta_t^F &= \max\left\{ 10^{-3},\ .03 Q_{.9}\left(\left\{ |\sqrt{\bar{y}_s^{(d)}} - \sqrt{b_s^\alpha}| : s \in \mathcal{F}_{t,20}^{(d)},\ \alpha \in \mathcal{A} \right\} \right) \right\}, \\ \eta_t^S &= .003 Q_{.9}\left(\left\{ |\sqrt{\bar{y}_s^{(d)}} - \sqrt{b_s^\alpha}| : s \in \mathcal{F}_t^{(d)},\ \alpha \in \mathcal{A} \right\} \right). \end{aligned} \tag{68}$$

Both rates are zero before the first release, pool absolute base errors over levels, and are recomputed as feedback arrives. The fast and slow hidden states $\tilde{\theta}_t^F, \tilde{\theta}_t^S$ and forecast are

$$\begin{aligned} \tilde{\theta}_t^{F,\alpha} &= .9\tilde{\theta}_{t-1}^{F,\alpha} + \eta_t^F \epsilon_t^\alpha, & \tilde{\theta}_t^{S,\alpha} &= \tilde{\theta}_{t-1}^{S,\alpha} + \eta_t^S \epsilon_t^\alpha, \\ \xi_t &= \Pi_{\mathcal{K}}(\sqrt{b_t} + \tilde{\theta}_t^F + \tilde{\theta}_t^S), & q_t^\alpha &= (\max\{\xi_t^\alpha, 0\})^2. \end{aligned} \tag{69}$$

The fast correction loses 10% per processed reference week, even without feedback; gaps do not compound decay. The slow correction persists. Re-runs reconstruct both rates and states. These rolling scales are separate from the frozen S_0.

Constant square-root rate: `sqrt_constant`. The flu production benchmark uses $\chi(x) = \sqrt{x/c_{\text{pop}}}$, where c_{pop} always equals population/100,000. Its update and forecast are

$$\begin{aligned} \tilde{\theta}_t^\alpha &= \tilde{\theta}_{t-1}^\alpha + .018\epsilon_t^\alpha, \\ \xi_t &= \Pi_{\mathcal{K}}\left(\sqrt{b_t/c_{\text{pop}}} + \tilde{\theta}_t\right), & q_t^\alpha &= c_{\text{pop}}(\max\{\xi_t^\alpha, 0\})^2. \end{aligned} \tag{70}$$

The rate .018 is in square-root counts-per-100,000 units. There is no adaptive rate, decay or second state. Delphi selected this constant using flu seasons 2024/25 and 2025/26, which overlap our evaluation period. Although we do not retune it, this is not an independently selected held-out benchmark for those seasons.

Zero boundary and plots. Square-root methods return zero when $\xi_t^\alpha < 0$ but learn from untruncated ξ_s^α. At zero outcomes, learning coverage can differ from coverage of the returned forecast. Both appear under "Single trackers" in script 04 and in the evaluation tree of script 06.

4 Guide to the outputs

The workflow writes plots under `output/pdf/` and score tables under `output/results/`. The active data design is `hub_first_months`; evaluation target weeks run from June 2025 through September 2026. Use the evaluation definitions in Section 3.4 when comparing plots and tables.

4.1 Fixed rates, coverage and loss

Script 02 explores the fixed-rate grid before comparing method families. `multiqt_error_tradeoff.pdf` shows calibration error against quantile loss by disease, feedback design, location and horizon. `multiqt_phase_rate_tradeoffs.pdf` shows how both measures change with the rate, separately overall, up to peaks and after peaks. Read them together: a reduction in calibration error need not reduce loss. Script 02 labels the rate multiplier κ as m.

4.2 Comparing the current method families

Script 04 uses the methods in Section 3.5; script 05 selects their settings as described in Section 3.3. The following comparisons answer different questions:

- **Median or full base.** Compare `studentized_base` with `studentized_full` to assess rebuilding around the median versus retaining every base quantile, both with a common base-width scale. Use the residual-RMS and constant-scale variants to examine the scale choice, and the zero-start variants to examine initialization.
- **Expert weighting.** Compare equal and learned mixtures within the same bank. Rate-grid methods also compare per-level and common weights, sharing, and an outer MultiQT correction. Weekly-start methods compare all starts with the eight most recent starts.

- **Feedback.** Compare delay 0, delay 1 and weekly re-runs within each method. Settings are tuned separately by feedback design, so the differences include both report timing and parameter selection.

`multiqt_peak_reasoning_tradeoff.pdf` compares methods by loss and calibration error. `multiqt_evaluation_tree.pdf` ranks ten methods per page by pooled after-peak loss; consult the trade-off plot for the full set. The US is separate from summaries over non-US locations. Forecast curves in `multiqt_method_forecasts/` and expert weights in `multiqt_weight_timelines/` explain individual cases. Filenames identify disease, horizon and feedback design, for example `covid_horizon2_delay0.pdf`. Each forecast file groups pages by location and method family. Weight pages show the expert weights, expert forecasts, mixture and latest counts.

For horizon 2 and delay 0, studentized forecast curves appear in `multiqt_method_forecasts/covid_horizon2_delay0.pdf`, pp. 5–6 (US), and `multiqt_method_forecasts/flu_horizon2_delay0.pdf`, pp. 5–6 and 11–12 (US and California). COVID rate-grid forecasts appear on p. 2 of the same forecast file, and their weights on p. 1 of `multiqt_weight_timelines/covid_horizon2_delay0.pdf`.

4.3 Base forecast paths and the three indicators

`multiqt_base_forecast_paths.pdf`, pp. 1–2, shows COVID and flu. Each location has a full latest-count timeline, all peaks retained by script 01, and blue median curves for horizons $-1, \ldots, 3$. Curves use reference weeks $-7, \ldots, 8$ around each peak and are trimmed to the same target-date windows. The three rows underneath show the three indicators at their reference dates, using information available by the corresponding cutoff. Peaks and latest counts are retrospective; neither enters these indicators.

How the indicators are computed. Use $Z(t, h; v)$ from the target-signal definition in (63), and threshold $\varepsilon = \log(1.05)$. Here date differences are calendar days: $t - 7$ means exactly one week earlier, not the previous retained round. Blue cells mean indicator 1, white 0 and gray unavailable. Each indicator is computed separately.

1. **Path: a decline within the current forecast curve.** The indicator is

$$I_t^{\text{path}} = \mathbf{1}\left\{ \max_{\substack{h=-1,\ldots,2 \\ Z_{t,h},\, Z_{t,h+1} \text{ available}}} (Z_{t,h} - Z_{t,h+1}) > \varepsilon \right\}.$$

   Horizons $0, 1, 2, 3$ must be available; include the -1 to 0 comparison only when the nowcast exists. This detects any sufficiently large adjacent decline, rather than comparing one target with all earlier horizons.

2. **Nowcast: a decline or slowing positive growth.** Set $z_0 = Z(t, -1; \iota_t)$, $z_1 = Z(t - 7, -1; \iota_{t-7})$ and $z_2 = Z(t - 14, -1; \iota_{t-7})$, and let $g = z_0 - z_1$ and $g_{\text{prev}} = z_1 - z_2$. Then

$$I_t^{\text{nowcast}} = \mathbf{1}\{g < -\varepsilon \text{ or } (g_{\text{prev}} > \varepsilon \text{ and } g_{\text{prev}} - g > \varepsilon)\}.$$

   Thus slowing growth can trigger the indicator before the nowcast falls. It requires three consecutive reference weeks with usable nowcasts.

3. **Revision: downward forecast revisions that become more negative.** For older horizons $j = 0, \ldots, 3$, compare each nowcast with forecasts for the *same target week*:

$$r_j = z_0 - Z(t - 7(j + 1), j; \iota_t),$$
$$r_j^{\text{prev}} = z_1 - Z(t - 7(j + 2), j; \iota_{t-7}).$$

Let J contain horizons with both comparisons available; require $|J| \geq 2$. The indicator is

$$I_t^{\text{revision}} = \mathbf{1}\Big\{\text{median}_{j\in J}\, r_j < -\varepsilon \text{ and } \text{median}_{j\in J}(r_j - r_j^{\text{prev}}) < -\varepsilon\Big\}.$$

The prior comparison uses its prior cutoff. These are revisions between *forecasts*, not revisions of reported hospital counts.

The logarithm is unnecessary for the current zero-threshold binary $c_{t,h}^{\text{ctx}}$, but it defines the relative-change scale of these three indicators and their nonzero threshold. They are diagnostic warnings, not probabilities of having passed a true peak. Decline uses the target-specific context defined in (63).

Why indicators are unavailable. A required median must exist at the exact reference date and horizon, be finite and nonnegative, be available by the cutoff and not be filled from an earlier forecast. Flu has no submitted forecasts for long summer intervals, although reported counts continue. Missing or removed rounds also interrupt the history: the nowcast indicator needs three consecutive nowcasts, and revision needs at least two matched older-horizon comparisons in both the current and prior round. Consequently, these two indicators can remain gray after forecasts resume, while Path is already available. For example, the unusable COVID reference weeks 25 January and 1 February 2025 leave Nowcast unavailable through 15 February and Revision through 22 February. Insufficient history at the start also gives gray cells. At the right edge, longer horizons can be absent from the prepared data because their finalized outcomes are not yet available; the indicators themselves do not use these outcomes. A gray cell does not mean indicator 0.

Reading the plot references. Paths in this section are relative to `output/pdf/`. The all-location trade-off markers average location scores equally; tree reductions use week-weighted pooling. Check horizon and feedback panels as well as pooled results.

References

[1] Tiffany Ding, Isaac Gibbs, and Ryan J. Tibshirani. *Calibrated Multi-Level Quantile Forecasting.* arXiv:2512.23671v1, 2025. MultiQT paper.

[2] Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. *Conformal PID Control for Time Series Prediction.* Advances in Neural Information Processing Systems 36, 2023. NeurIPS paper.

[3] CMU Delphi. *Delphi hub forecasting: calibration experiments and tracker code.* Pinned source revision `9abebcfb`, inspected 9–10 November 2026. E09: constant learning rate and revision delay; E15: median shift and width change; tracker implementation.